

Olist E-Commerce Analysis

A SQL Portfolio Project on Brazil's Online Marketplace

Author : Zulfah Adams

Dataset : Olist Brazilian E-Commerce Public Dataset (Kaggle)

Database : PostgreSQL

Date: 2025

Abstract

This project uses SQL to analyze the Olist Brazilian e-commerce dataset, looking at five areas: who's selling on the platform and where the market is based, how reliably orders get delivered, how location affects shipping cost and speed, what's driving revenue, and whether customers come back to buy again. Eight queries were written against a PostgreSQL database, ranging from simple grouping and counting to window functions and data based calculations.

A consistent pattern in almost every query : customers and sellers in Brazil's northern states are consistently worse off than those in the southeast. They deal with higher freight costs, longer delivery times, and fewer orders overall. This wasn't something I set out to find, but rather a recurring pattern.

Research Questions

This project is structured around five business areas, each addressed by one or more SQL queries. The central questions guiding the analysis are:

Market Concentration:

Which sellers and states account for the bulk of orders? Is everything concentrated in one state?

Operational Health:

How often do orders run late, and are delays spread evenly across the country or concentrated in certain regions ?

Logistics Performance:

Does where a customer lives affect how much they pay for shipping and how long they wait, and are northern states getting a worse deal?

Revenue Drivers:

Which product categories bring in the most revenue, and how do customers prefer to pay?

Customer Loyalty:

How many customers come back for a second order, and what does that pattern look like across the platform?

Introduction

Brazil's e-commerce market is huge, but it's far from evenly spread. A handful of states, mostly in the southeast account for most of the orders, while large parts of the country, particularly the north, are harder and more expensive to reach. The Olist dataset gives a good window into this. It covers around 99000+ orders placed on the Olist marketplace between 2016 and 2018, with data on orders, products, customers, payments and order items.

I built this project to practise and demonstrate SQL skills that come up a lot in analyst roles: joins across multiple tables, window functions, grouping and aggregation, and working with dates in PostgreSQL. The eight queries are grouped into the five areas above, which are market concentration, operational health, logistics, revenue, and customer loyalty.

The clearest pattern across the whole project is geographic: States like Roraima, Amapa, and Acre in the north consistently come last, fewer orders, longer delivery times, higher freight costs. While São Paulo, Rio de Janeiro, and Minas Gerais dominates almost every metric. It shows up independently across the logistics, revenue, and market concentration queries, which suggests its a structural issue rather than a singular occurrence.

Methodology:

Dataset

The Olist Brazilian e-commerce dataset is publicly available on Kaggle and comes as nine separate CSV files covering orders, customers, sellers, products, payments, reviews, and geolocation. I loaded five relevant files to the analysis into a PostgreSQL database without changing the underlying data.

Query Design

Each of the eight queries answers a specific question and is written to stand on its own, someone reading the SQL file shouldn't need the rest of the project to follow what it's doing. Across the queries I used aggregate functions (COUNT, SUM, AVG), GROUP BY with HAVING, multi-table joins, window functions like CTEs and subqueries, and PostgreSQL's `::date` casting for working out time differences.

Data Quality

While writing the logistics query, a negative output kept returning. I confirmed that the carrier delivered date was stored as a later date than the customer delivered date in the dataset, certainly a clear mislabelment. Columns were swapped in the query to produce logically correct positive transit times. No other filtering was applied anywhere else in the project, and freight and revenue figures were taken as they appear in the dataset.

Scope and Limitations

This is a portfolio project, not a research paper. The goal was to show practical SQL skills on a real dataset, not to run a formal study. The findings below are descriptive rather than statistically tested. The dataset only covers 2016 to 2018, so some patterns especially over a longer window. Figures aren't adjusted for inflation.

Initial Analysis - Results and Discussion

Q1 - Market Concentration

Looking at where orders come from across Brazil, Sao Paulo accounts for roughly 42% of all orders placed on the platform. When we add Rio de Janeiro and Minas Gerais, the top three states account for around two thirds of total order volume. Everything else is split across the remaining 24 states, most of them contributing in the low single digits. This makes sense given how population and economic activity are distributed in Brazil, but it does mean the business is fairly exposed if anything were to shift in that region such as a local economic downturn, a competitor opening a regional distribution center, or even a logistics disruption could have an extreme impact on overall revenue and conversely it points to states in the north and northeast as relatively untapped. One thing worth noting is that this doesn't account for population size, so we can't say whether SP is overperforming relative to its size or just reflects where most of Brazil's online shoppers actually live.

Q2: Operational Health

About 97% of orders are successfully delivered, with cancellations and "unavailable" orders each under 1%. That's a healthy completion rate overall. The main gap here is that the data doesn't explain why orders fail, whether it's customer driven, seller driven or something else. So while the numbers look good, there's not much to act on without more context behind those failures.

Q3: Logistics Performance

Delivery times differ massively by region. Customers in Sao Paulo typically wait 5-6 days for carrier transit, while customers in Roraima and Amapa wait closer to 24 days which is 4 to 5 times longer. Freight costs rise in line with this. It raises the question of whether some kind of regional fulfillment presence outside the Southeast could help, though this only looks at carrier transit time and not total processing time, so the full delivery experience is likely a bit longer across the board.

Q4: Delivery Promise Accuracy

Most states have decent on time delivery rates, but Alagoas and Maranhao fall behind at around 76 to 80 percent. Interestingly, some of the slowest states in absolute terms like Acre and Rondonia actually deliver well ahead of their estimated dates, sometimes by close to 20 days. This suggests delivery estimates for remote areas are heavily padded, which protects against missed promises but might also be discouraging customers by setting expectations unnecessarily low. Also, this analysis only checks if an order was on time or late, it doesn't show how late so a one day delay and a two week delay would look the same in the numbers.

Q5: Revenue Drivers

Beauty and health products bring in the most revenue at around R\$1.26 million, ahead of categories like watches & gifts and bed/bath/table. Watches & gifts stand out for having the highest average item price at roughly R\$200. These numbers are based on item price only, not profit margin or freight cost, so the highest revenue doesn't necessarily mean most profitable - but it's still a useful signal for which categories matter most to the platform.

Q6: Customer Loyalty

This was probably the most interesting result: around 97% of customers have only ever placed one order. Repeat purchasing is essentially. It's possible this partly reflects the kinds of products being sold(furniture,electronics,etc. aren't things people buy often), but it could also point to genuine retention issues tied to long delivery times seen in some regions. The dataset only covers about two years, so some "one-time" customers may simply not have had the chance to return yet.

Q7: Payment Behaviour

Credit cards are the dominant payment method at roughly 75% of orders with boleto(a Brazilian bank slip payment) making up about 19%. That's a good portion of customers who likely don't use or don't trust online card payments, a reminder that payment flexibility plays a real role in who can access the platform. Voucher based orders had a noticeably lower average value (R\$65), suggesting they're tied to smaller,discount driven purchases.

Q8: Freight Cost vs Late Delivery

As freight cost increases, so does both transit time and likelihood of late delivery. Higher shipping costs don't appear to buy any kind of speed or reliability, if anything it does the opposite. This fits a broader pattern across the analysis, freight cost is mostly a proxy for distance and logistical difficulty, meaning customers further from the southeast are paying more and waiting longer. This is based on freight cost tiers rather than distance directly, so it's a bit of an indirect measure, but the pattern is consistent enough to be meaningful.

Cross - Query Theme: Geographic Disadvantage

The pattern that ties most of this together is geography. Northern and northeastern states show up with fewer orders in the market concentration query, more delivery delays in the operational health query, longer transit times and higher freight costs in the logistic query. The fact that the same region keeps appearing across completely separate queries suggests this isn't down to any one category or seller, it speaks to a bigger infrastructure gap between Brazil's regions.

Summary and Conclusion

This project used SQL across five business areas to look at the Olist Brazilian e-commerce dataset, producing eight queries that cover market structure, logistics, revenue, and customer behaviour.

Overall, the platform looks operationally solid, high delivery success rates, a dominant and reliable payment method, and clear revenue leaders, but with two issues that shows up consistently across the dataset: geography and retention.

Customers in the south and southeast get a noticeable better deal, faster deliveries, lower freight costs, and more accurate delivery estimates - while customers further north wait longer, pay more and get overly cautious delivery windows. This pattern shows up across almost every query, making it the most interesting finding overall, and it points to both a problem and an opportunity. The other big issue is retention around 97% of customers only order once, leaving room for a lot of improvement.

On the technical side, this project covers multiple joins, window functions, date arithmetic in PostgreSQL, and basic data cleaning. All 8 queries are available on Github, linked below.

Appendix

All SQL files for this project are in the Github repository linked from my CV. The repository README has a short write up of each queries findings with links to the individual SQL files.

Query Index

Q1 (Market Concentration)

Q2 (Operational Health)

Q3 (Logistics Performance)

Q4 (On time accuracy)

Q5 (Revenue Drivers)

Q6 (Customer Loyalty)

Q7 (Payment Behaviour)

Q8 (Freight vs Speed)

Dataset Schema

Olist Brazilian E-Commerce Public Dataset:

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>